Highly accurate protein structure prediction with AlphaFold

Jumper et al., *Nature* 596 (2021) [JEP+21]

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`github.com/BenBullinger/ML4Genomics-Presentation-AF2`

Machine Learning for Genomics (263-5351-00L), ETH Zürich, Fall 2026

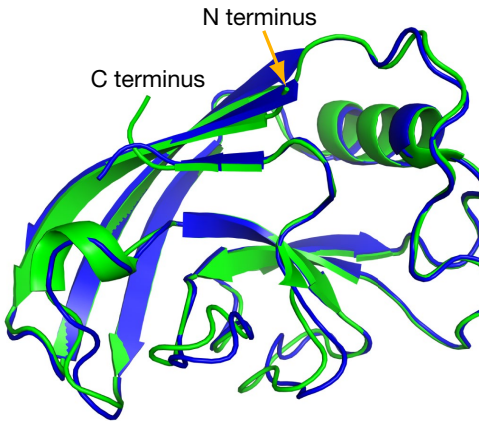
The problem: sequence to structure

SIFSYITESTGTPS...

A chain of amino acids (residues), each with the same backbone and its own side chain

Billions of sequences are known, about 100,000 structures (Protein Data Bank, PDB).

Predict the 3D position of every atom except hydrogen from the sequence.

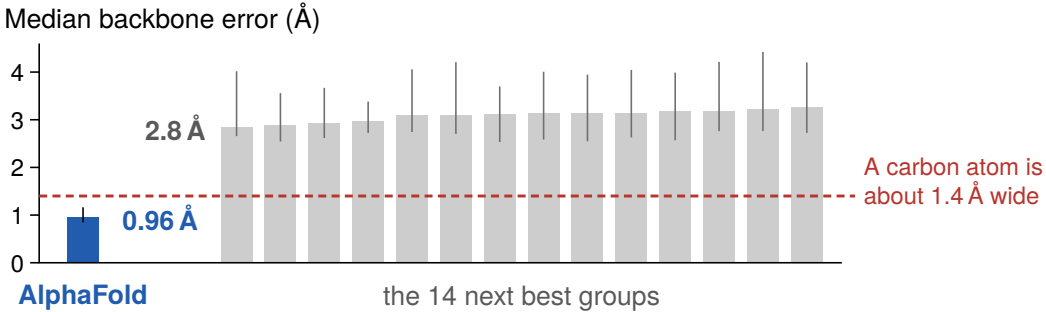


Experiment in green, prediction in blue

[JEP+21], Fig. 1b. PDB 6Y4F

The benchmark: CASP14

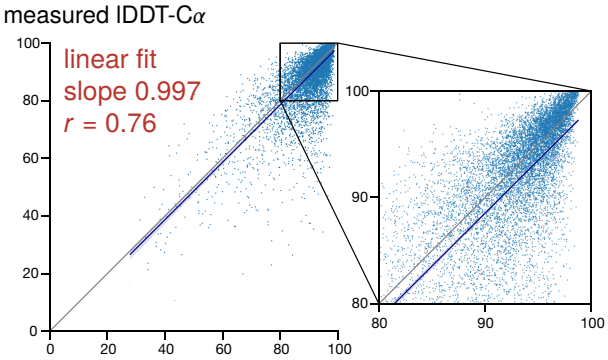
CASP14, the Critical Assessment of protein Structure Prediction, 2020. A blind test on 87 domains, the parts of a protein that fold on their own.



The median backbone error is 0.96 Å, less than the width of a carbon atom.

Redrawn from [JEP+21], Fig. 1a. Median Cα r.m.s.d.$_{95}$ per group, 95 % confidence intervals

Confidence: pLDDT against measured lDDT

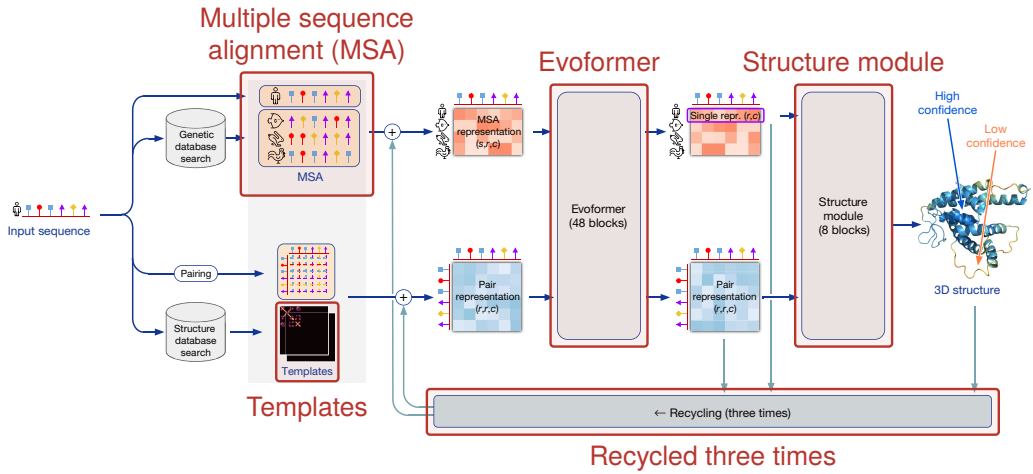


The local distance difference test (lDDT) scores a structure from 0 to 100. The network's own prediction of it, per residue, is pLDDT.

Predicted and measured lDDT agree, slope 1.0, $r = 0.76$**.**

[JEP+21], Fig. 2c, n = 10,795 chains. Trained on the lDDT of its own predictions, SI 1.9.6

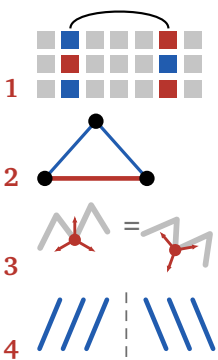
The network: inputs, Evoformer, structure module



The network outputs the 3D coordinates directly.

[JEP+21], Fig. 1e

The design: four constraints built into the network 1 2 3 4

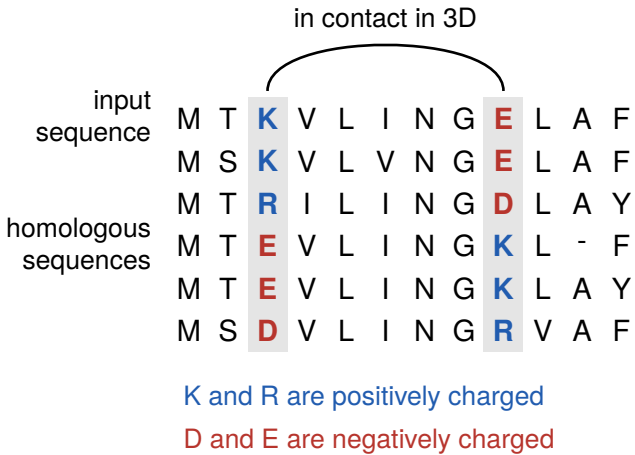


	Known before training	Built into AlphaFold 2
1	Residues in contact change together across homologous proteins	Attention over the MSA of homologous proteins
2	Distances must fit one 3D structure	Pair features updated through triangles
3	A rotation or translation changes nothing	A coordinate system per residue, attention over distances
4	A protein and its mirror image differ (chirality)	A loss that compares atoms residue by residue

Each constraint is built into one part of the network.

[JEP+21], p. 584: “evolutionary, physical and geometric constraints”

The input: the multiple sequence alignment

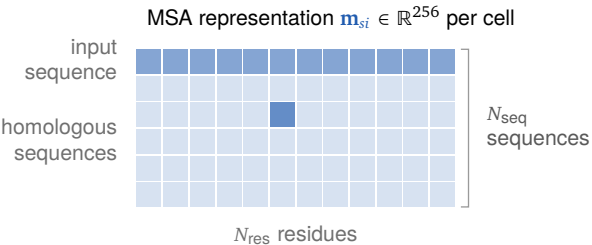


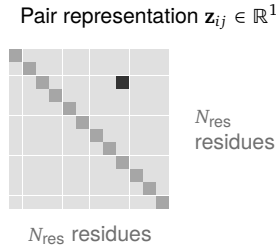
Rows are homologous proteins, columns aligned positions.

The alignment itself is the input to the network.

Illustration after [MCS+11]

The representations: MSA array and pair array



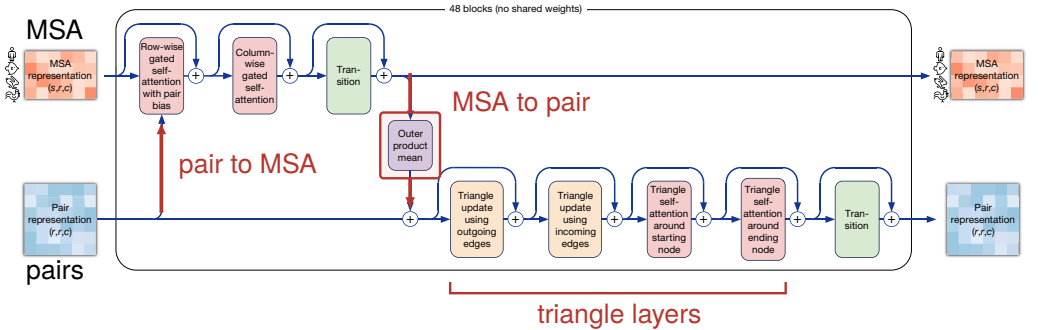


One vector per alignment cell, one per ordered pair of residues.

Each of the 48 Evoformer blocks updates these two arrays.

Evoformer: one block

1. Residues in contact change together across homologous proteins.

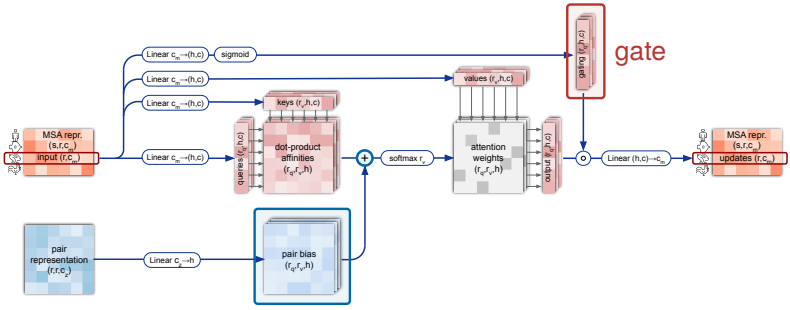


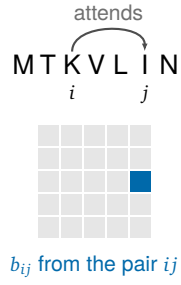
Attention along the rows and the columns. Each array updates the other.

Every residue pair is an edge. The network infers which pairs are in contact.

Evoformer: row attention with pair bias

1. Residues in contact change together across homologous proteins.



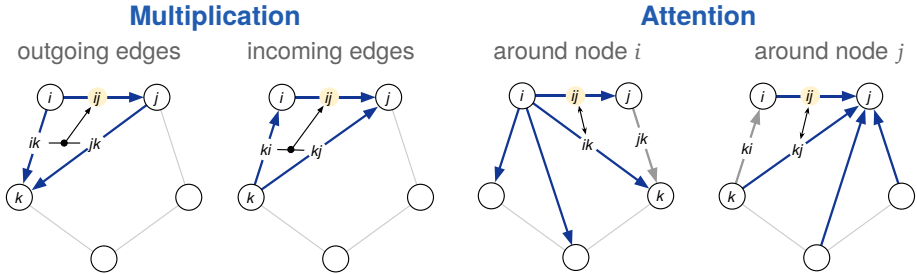


Residue i attends to j in its row, biased by the pair ij. $a_{sij} = \mathrm{softmax}_j\left(\frac{1}{\sqrt{c}}\,\mathbf{q}_{si}^{\top}\mathbf{k}_{sj} + b_{ij}\right)$

The pair representation biases the attention within every sequence.

Evoformer: triangle updates

2. Distances must satisfy the triangle inequality.



Edge ij is updated from the edges ik and jk, for every third residue k.

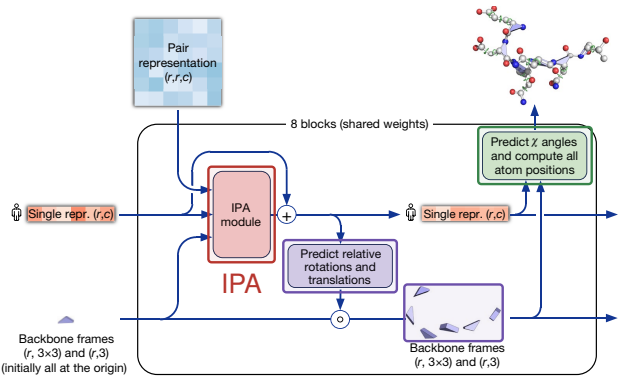
$$\tilde{\mathbf{z}}_{ij} \propto \sum_k \mathbf{a}_{ik} \odot \mathbf{b}_{jk}$$

Each pair update uses the two other edges of its triangle.

[JEP+21], Fig. 3c. SI Algorithms 11 to 14

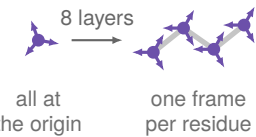
Structure module: residues as rigid bodies

3. A structure is the same object after any rotation or translation.



A frame T_i is one rotation and one translation per residue.

In each of 8 layers invariant point attention (IPA) updates the residue features and every frame is moved. Then the atoms are placed from torsion angles.

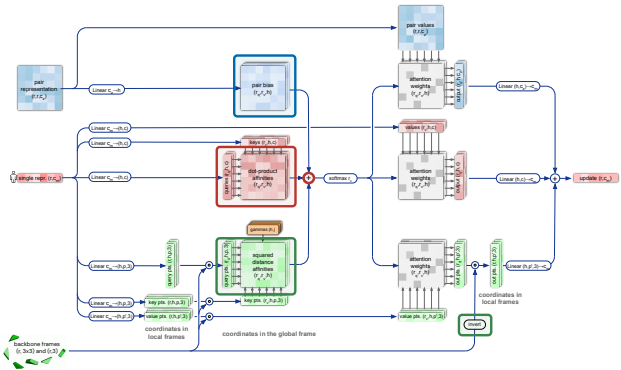


Each residue is moved as a free rigid body.

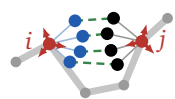
[JEP+21], Fig. 3d. SI Algorithm 20

Structure module: invariant point attention (IPA)

3. A structure is the same object after any rotation or translation.



Each residue predicts points in its own frame. The attention logit is the dot product, plus the pair bias, minus the squared distance between the points of i and j.



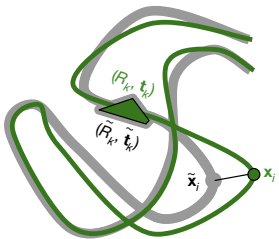
$$\mathbf{q}_i^\top \mathbf{k}_j + b_{ij} - \gamma \sum_p \| T_i \circ \vec{\mathbf{q}}_{ip} - T_j \circ \vec{\mathbf{k}}_{jp} \|^2$$

The attention weights are invariant to a global rotation and translation.

SI 1.8.2, Suppl. Fig. 8, Algorithm 22. Invariance proof on backup slide 1

The loss: frame aligned point error (FAPE)

4. A protein and its mirror image are different molecules.



Green predicted, grey true

Every atom j is expressed in the coordinates of every frame k, in the prediction and in the experiment. d_{kj} is the distance between the two local positions.

$$\mathcal{L}_{\text{FAPE}} = \frac{1}{Z} \text{mean}_{k,j} \min(d_{\text{clamp}}, d_{kj}), \text{ with}$$
$$Z = d_{\text{clamp}} = 10\,\text{Å}$$

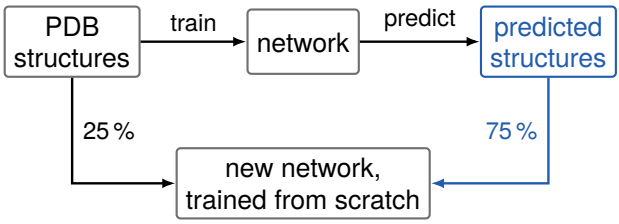


atom j in frame k

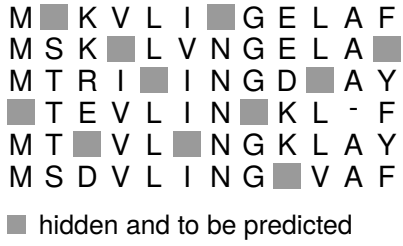
FAPE is invariant to rotations and translations, but not to reflection.

Training: self-distillation, masked MSA, total loss

Self-distillation adds structures predicted for 350,000 sequences.



A masked-MSA loss hides 15 % of the alignment entries.



The loss is a weighted sum of five terms (SI eq. 7).

0.5 FAPE on the final structure
0.5 FAPE after each of the 8 layers, Cα only
0.3 residue distances in bins, from the pair array
2.0 the hidden alignment entries
0.01 pLDDT against the lDDT reached

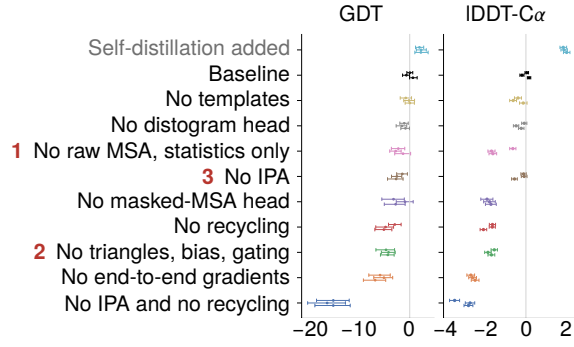
Training also uses sequences without a known structure.

SI 1.3, 1.9, eq. 7. Self-distillation after [XLHL20], masking after [DCLT19]

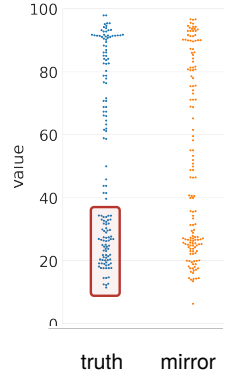
Ablations: accuracy lost without each component

1 2 3 4

An ablation removes one component and retrains



Trained without FAPE, GDT against

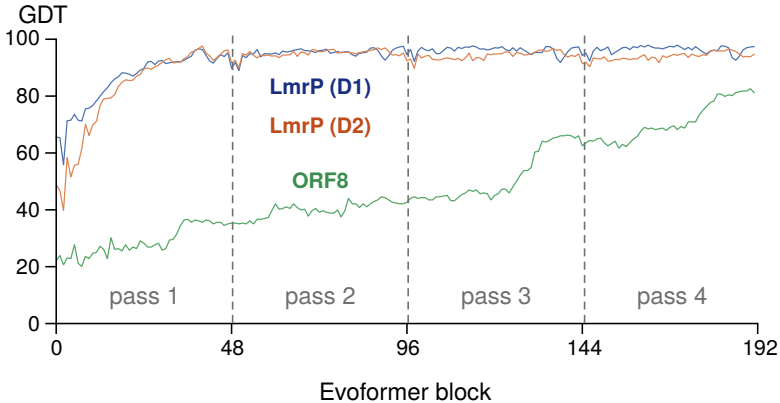


4 Without FAPE, the low group matches the mirror image (SI 1.9.3).

Removing any of the four components lowers the accuracy on CASP14.

[JEP+21], Fig. 4a and Suppl. Fig. 9, labels re-set. GDT (global distance test, 0 to 100) on CASP14 domains, lDDT-Cα on PDB chains. SI 1.9.3

Interpretation: intermediate structures in the Evoformer

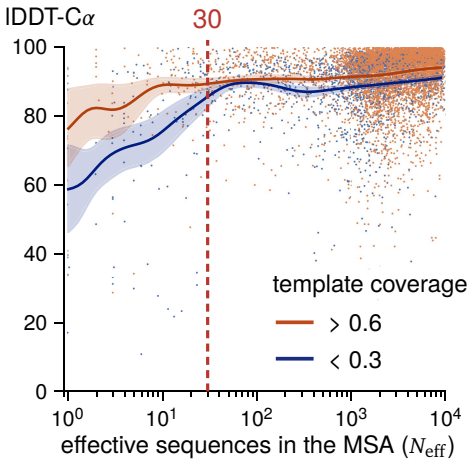


One structure module trained after every Evoformer block, the Evoformer frozen.

LmrP reaches its final structure after a few blocks, ORF8 only near the end.

[JEP+21], Fig. 4b. LmrP: CASP14 target T1024. ORF8: T1064, SARS-CoV-2

Limitations: MSA depth and complexes



Below about 30 related sequences in the alignment, accuracy drops sharply. A template of a related structure makes up for part of it.

Chains whose contacts are mostly with other chains are predicted worse. The network is trained on single chains.

Accuracy depends on how many related sequences are known.

[JEP+21], Fig. 5a. N_{eff}: sequences after merging near-duplicates

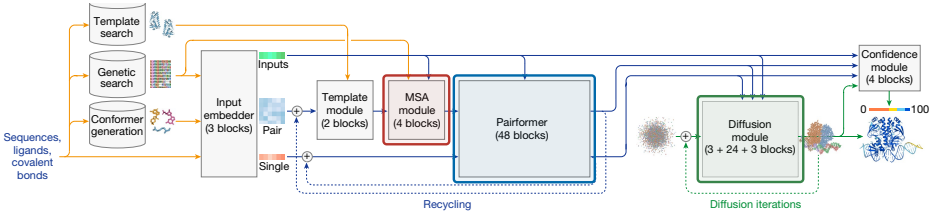
After AlphaFold 2: 2021 to 2025

2021	RoseTTAFold is published the same day	[BDA+21]
	Structures predicted for 98.5 % of human proteins	[TAW+21]
	AlphaFold-Multimer predicts complexes	[EOP+21]
2022	ESMFold predicts from one sequence alone	[LAR+23]
2024	AlphaFold 3: proteins, DNA, RNA and ligands (small molecules)	[AAD+24]
	Nobel Prize in Chemistry for Baker, Hassabis and Jumper	[Roy24]
2025	AlphaGenome predicts gene regulation from DNA	[ALC+26]

Multimer and AlphaFold 3 predict complexes, ESMFold needs no alignment.

AlphaFold 3: the four constraints revisited

1 2 3 4



1 The MSA module shrinks to four blocks that write the alignment into the pairs.

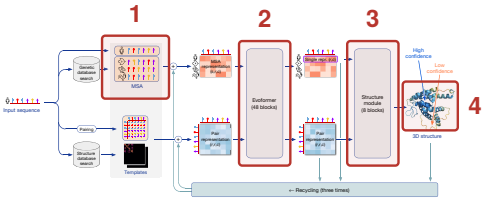
2 The Pairformer keeps the 48 blocks of triangle updates.

3 4 Diffusion replaces IPA and FAPE. Chirality is wrong in 4.4 % of cases (p. 497).

Two constraints were dropped and learned from data. The triangles were kept.

[AAD+24], Fig. 1d, p. 494 to 497. The 4.4 % is the chirality violation rate on the PoseBusters ligand benchmark, p. 497. Our reading follows [Sut19]

Summary: the four constraints and their ablation



		Without it
1	attention over the alignment	−2.2 GDT
2	triangles on the pairs	−4.6 GDT
3	IPA in the structure module	−2.3 GDT
4	FAPE in local frames	mirror images

Every constraint improves the accuracy in the ablation. AlphaFold 3 kept only the triangles.

Questions for the audience

1 The confidence pLDDT is trained on the model's own errors. When would you distrust it?

2 Accuracy falls below about 30 related sequences. Has the network learned the physics of folding, or the statistics of evolution?

3 AlphaFold 2 builds symmetry and chirality into the network. AlphaFold 3 learns them from data. When is a built-in constraint worth it?

github.com/BenBullinger/ML4Genomics-Presentation-AF2

References: structure prediction

[JEP+21] J. Jumper, R. Evans, A. Pritzel, et al., "Highly accurate protein structure prediction with AlphaFold," *Nature* 596, 583 to 589 (2021). CC BY 4.0.

[TAW+21] K. Tunyasuvunakool, J. Adler, Z. Wu, et al., "Highly accurate protein structure prediction for the human proteome," *Nature* 596, 590 to 596 (2021).

[BDA+21] M. Baek, F. DiMaio, I. Anishchenko, et al., "Accurate prediction of protein structures and interactions using a three-track neural network," *Science* 373, 871 to 876 (2021).

[EOP+21] R. Evans, M. O'Neill, A. Pritzel, et al., "Protein complex prediction with AlphaFold-Multimer," *bioRxiv* 2021.10.04.463034 (2021).

[LAR+23] Z. Lin, H. Akin, R. Rao, et al., "Evolutionary-scale prediction of atomic-level protein structure with a language model," *Science* 379, 1123 to 1130 (2023). Preprint: *bioRxiv* 2022.07.20.500902.

[AAD+24] J. Abramson, J. Adler, J. Dunger, et al., "Accurate structure prediction of biomolecular interactions with AlphaFold 3," *Nature* 630, 493 to 500 (2024). CC BY 4.0.

[MCS+11] D. S. Marks, L. J. Colwell, R. Sheridan, et al., "Protein 3D structure computed from evolutionary sequence variation," *PLoS ONE* 6, e28766 (2011).

References: methods and context

[ALC+26] Ž. Avsec, N. Latysheva, J. Cheng, et al., “Advancing regulatory variant effect prediction with AlphaGenome,” *Nature* 649, 1206 to 1218 (2026), doi 10.1038/s41586-025-10014-0. Preprint 2025.

[DCLT19] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” NAACL-HLT (2019).

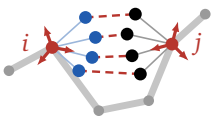
[XLHL20] Q. Xie, M.-T. Luong, E. Hovy, Q. V. Le, “Self-training with noisy student improves ImageNet classification,” CVPR (2020).

[Sut19] R. Sutton, “The Bitter Lesson,” essay, incompleteideas.net (2019).

[Roy24] The Royal Swedish Academy of Sciences, “Scientific background to the Nobel Prize in Chemistry 2024” (2024).

IPA: invariance and equivariance

SI 1.8.2 and 1.8.3, equations 3 to 6



A global motion T_{global} cancels in every distance between the query points and the key points placed by the frames.

$$\left\| T_{\text{global}} \circ T_i \circ \vec{q}_i - T_{\text{global}} \circ T_j \circ \vec{k}_j \right\| = \left\| T_i \circ \vec{q}_i - T_j \circ \vec{k}_j \right\|$$

The output points are mapped back into frame i, so T_{global} cancels against its inverse.

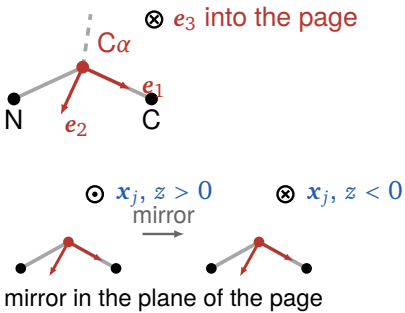
$$(T_{\text{global}} \circ T_i)^{-1} \circ \sum_j a_{ij}\, T_{\text{global}} \circ T_j \circ \vec{v}_j = {T_i}^{-1} \circ \sum_j a_{ij}\, T_j \circ \vec{v}_j$$

The frame update is composed in the frame's own coordinates, $(T_{\text{global}} \circ T_i) \circ T_i^{\text{upd}} = T_{\text{global}} \circ (T_i \circ T_i^{\text{upd}})$.

Exact equivariance under rotations and translations. A reflection is not covered, since frames built from atoms do not reflect with them.

Chirality: where the handedness comes from

SI 1.8.1 and 1.9.3, Algorithm 21, equations 11 to 17



Frames are built from N, Cα and C by Gram–Schmidt with $\boldsymbol{e}_3 = \boldsymbol{e}_1 \times \boldsymbol{e}_2$, always right-handed.

Reflect all atoms. The rebuilt frames are not the reflected frames. In local coordinates exactly z changes sign, so the unclamped error is

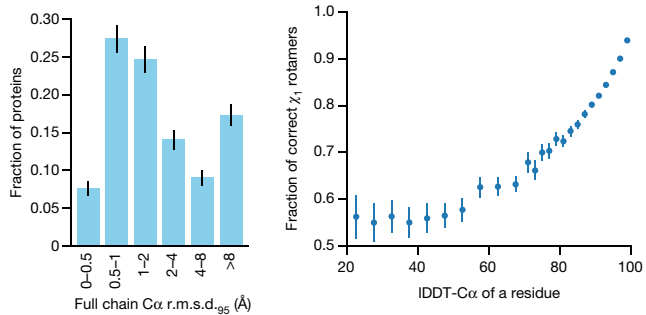
$$\mathrm{FAPE}(X, X_{\mathrm{mirror}}) = 2 \sum_{ij} \left| (T_i^{-1} \circ \boldsymbol{x}_j)_z \right|$$

Trained with a distance-only loss (dRMSD) instead, the model often predicts the mirror image (Suppl. Fig. 9).

FAPE distinguishes the two hands, the training data select the right one.

Generalisation: recent PDB structures

Fig. 2a and 2b, recent PDB structures



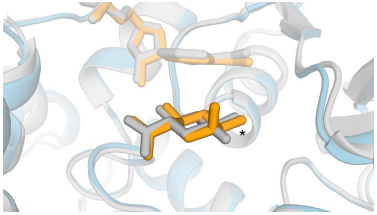
The 3,144 chains were released after the training cutoff and have no close template. Side chains are correct where the backbone is correct.

Median backbone error 1.46 Å on new structures.

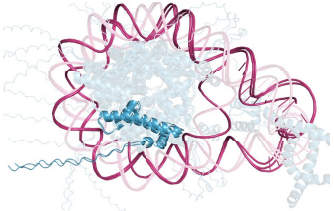
[JEP+21], Fig. 2a and 2b, p. 585

AlphaFold 3: the reported failures

[AAD+24], Fig. 5b and 5e



A sugar bound to an enzyme, PDB 7CTM. The predicted ligand (orange) has the wrong hand at the starred carbon, the experiment is grey.



A trinucleosome, PDB 7PEU. Two protein chains (blue) and the DNA (pink) run through each other in the prediction.

Both are penalised in the ranking. Both remain in the predictions.